

Erdős Problem #325

EP-325

L1 - TRACTABLE

PARTIALLY SOLVED

Number Theory

ORIGINAL PROBLEM STATEMENT

Let $k \geq 3$ and $f_{k,3}(x)$ denote the number of integers $\leq x$ which are the sum of three nonnegative k th powers. Is it true that $f_{k,3}(x) \gg x^{3/k}$ or even $\gg_{\epsilon} x^{3/k-\epsilon}$?

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Archive: <https://www.unsolvedmath.com/problems?difficulty=1>
Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>